Clément Hoffmann

Curriculum Vitae

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clement.hoffmann@hotmail.fr • hoffmanncl.github.io
Google Scholar 0000-0001-9744-1539 hoffmanncl

Research Summary

Postdoctoral Researcher at NTT Social Informatics Laboratories (Japan), specializing in the security of cryptographic implementations. My research lies at the intersection of side-channel analysis and lattice-based cryptography, with a core objective of bridging the gap between leakage theory and practical physical security. While my expertise includes concrete attacks and secured implementations, my primary focus is on establishing rigorous reduction proofs from physical leakage problems to classical learning problems. Additionally, I work on Boolean functions applied to Hybrid Homomorphic Encryption, as well as non-leakage lattice-based cryptography such as Updatable Public Key Encryption or Threshold Signatures.

Professional Experience & Education

NTT Social Informatics Laboratories **Tokyo, Japan**
Postdoctoral Researcher *2025–Present*

- Design of masking gadgets and secure implementations for lattice-based primitives, with a focus on discrete Gaussian sampling (HAWK, FrodoKEM) and ML-DSA.
- Concrete side-channel attacks on post-quantum implementations (template attacks, soft-analytical side-channel attacks) on ARM Cortex-M4 targets.
- Reductions from physical leakage assumptions (LWPR, LWE-OD) to standard lattice problems.
- Post-quantum threshold signatures: co-designed Kettle, a threshold version of HAWK submitted to the NIST Multi-Party Threshold Cryptography call.

UCLouvain **Louvain-la-Neuve, Belgium**
PhD in Cryptology *2020–2025*

- Supervised by **François-Xavier Standaert** and **Pierrick Méaux**.
- Defended my thesis, *Post-quantum side-channel security from hard physical learning problems*, in January 2025. Committee composed of Pr. François-Xavier Standaert, Dr. Pierrick Méaux, Dr. Melissa Rossi, Dr. Mehdi Tibouchi, Pr. Thomas Peters, Pr. David Bol.
- Thesis awarded a **Special Mention for Scientific Impact** at the ICTEAM Best Thesis Award (2026).
- Side-channel analysis and countermeasures for lattice-based schemes.
- Applied learning problems theory to side-channel analysis.
- Fully Homomorphic Encryption (FHE) and Hybrid Homomorphic Encryption (HHE).

Thales SIX GTS France — **Gennevilliers, France**
Research Intern (6 months) — *2020*

Homomorphic encryption implementation (TFHE-based transciphering), under the supervision of Thomas Ricosset and Pierrick Méaux. This work led to the Indocrypt 2020 publication.

University of Limoges — **Limoges, France**
Master degree in Mathematics and Cryptography — *2018–2020*

- Valedictorian.
- Master thesis in lattice & code-based cryptography: *Etude et comparaison de cryptosystèmes post-quantiques : McEliece - NTRUEncrypt*.

Sorbonne University (UPMC) — **Paris, France**
Bachelor degree in Mathematics — *2016–2018*

Specialization in algebra and theoretical computer science.

Awards & Academic Service

Awards:
- Special Mention for Scientific Impact, ICTEAM Best Thesis Award, UCLouvain (2026).
- Valedictorian, Master in Mathematics and Cryptography, University of Limoges (2020).

PC member: ISC 2026, CHES 2027.

Sub-reviewer: EUROCRYPT 2026 and 2027, CRYPTO 2026, CHES 2024 and 2026, PKC 2026 and 2027, IEEE EuroS&P 2026, IMACS 2026.

Teaching & Skills

Teaching:
- Teaching Assistant at UCLouvain for *LMAT2450 – Cryptography* (exercise sessions, in English), 2021–2024.
- Co-supervised the master thesis (TFE) of Thibaud Schoenauen at UCLouvain (2024), leading to the joint publication *Leveled Software Implementation of Polka and Comparison with Uniformly Masked Kyber* (ACNS 2025 Workshops).

Languages: French (Native), English (C1/C2), Japanese (JLPT N4/N3), Spanish (former B2).

Programming: C, C++, Rust, Python, SageMath.

Embedded: C and assembly development on microcontrollers, masked implementations, benchmarking, side-channel evaluation and attacks.

Peer-Reviewed Publications

SIDE-CHANNEL ANALYSIS, ATTACKS AND COUNTERMEASURES

Knuth–Yao Masked Sampling: Improved Performance for FrodoKEM's and HAWK's Gaussian Samplers

Authors: Calvin Abou Haidar, Clément Hoffmann

Venue: CHES 2027 (ePrint 2026/2287, code)

Summary: Presented the first masked evaluation of the Knuth–Yao sampler, showing that its tree structure, evaluated as a bitsliced Boolean circuit, requires exactly $L - 2$ masked AND gadgets (L being the total Hamming weight of the probability expansions). Provided a generic framework compiling any finite-precision distribution into an optimized masked C implementation, instantiated on all FrodoKEM and HAWK parameter sets: on ARM Cortex-M4, it outperforms the state-of-the-art masked CDT approach for every distribution and masking order, running up to twice as fast at order 1.

Slicing Bits and Cutting Costs in CDT Sampling: High-Order Masking of FrodoKEM's Gaussian Sampler, Revisited

Authors: Calvin Abou Haidar, Thomas Espitau, Clément Hoffmann, Mehdi Tibouchi

Venue: CHES 2026 (ePrint 2026/1363, code)

Summary: Optimized the masked CDT gadget for Gaussian sampling proposed by Eid et al. Demonstrated that algorithmic improvements combined with bitslicing techniques more than double the sampling throughput. Designed highly efficient gadgets tailored for low-depth settings and provided a public, open-source implementation targeting microcontrollers.

Maskaglia: A New, Efficient Approach to Masked Discrete Gaussian Sampling

Authors: Calvin Abou Haidar, Thomas Espitau, Clément Hoffmann, Mehdi Tibouchi

Venue: CRYPTO 2026 (ePrint 2026/988)

Summary: Introduced a novel rejection-based methodology for masked discrete Gaussian sampling. By combining this approach with advanced algorithmic optimizations and efficient implementation techniques, we achieved a performance speedup of approximately $20\times$ compared to state-of-the-art.

Leveled Software Implementation of Polka and Comparison with Uniformly Masked Kyber

Authors: Thibaud Schoenauen, Clément Hoffmann, Charles Momin, Thomas Peters, François-Xavier Standaert

Venue: ACNS 2025 Workshops (DOI)

Summary: Co-supervised the first leveled software implementation of the POLKA PKE scheme. Validated that POLKA provides significant performance gains starting from two shares, demonstrating the practical viability of leakage-resistant designs for post-quantum standards.

Protecting Dilithium against Leakage: Revisited Sensitivity Analysis and Improved Implementations

Authors: Melissa Azouaoui, Olivier Bronchain, Gaëtan Cassiers, Clément Hoffmann, Yulia Kuzovkova, Joost Renes, Tobias Schneider, Markus Schönauer, François-Xavier Standaert, Christine van Vredendaal

Venue: CHES 2023 (open access, code)

Summary: Optimized side-channel countermeasures for the ML-DSA (Dilithium) NIST standard. Conducted a refined sensitivity analysis and developed specialized masking gadgets. Greatly contributed to the high-performance implementation and benchmarking of protected Dilithium.

POLKA: Towards Leakage-Resistant Post-quantum CCA-Secure Public Key Encryption

Authors: Clément Hoffmann, Benoît Libert, Charles Momin, Thomas Peters, François-Xavier Standaert

Venue: PKC 2023 (ePrint 2022/873)

Summary: Co-designed POLKA, a leakage-resistant PKE scheme optimized for low-cost side-channel protections. Contributed to architectural features that eliminate the expensive Fujisaki-Okamoto re-encryption step and allow for masking overheads that scale linearly with the number of shares. Integrated key-homomorphic properties and analysis of hard physical learning problem to achieve native leakage resistance.

Systematic Study of Decryption and Re-encryption Leakage: The Case of Kyber

Authors: Melissa Azouaoui, Olivier Bronchain, Clément Hoffmann, Yulia Kuzovkova, Tobias Schneider, François-Xavier Standaert

Venue: COSADE 2022 (ePrint 2022/036)

Summary: Conducted a systematic and quantitative investigation of Kyber against SPA and DPA attacks. Developed shortcut formulas to evaluate attack strength versus noise levels and analyzed the high overhead of masking countermeasures. Evaluated the relative efficiency of side-channel attacks on CPA decryption versus FO re-encryption process.

LEARNING PROBLEMS REDUCTION AND PHYSICAL SECURITY

Hardness of Learning with Physical Rounding and Noise from Learning With Errors

Authors: Clément Hoffmann, Emeline Repel, Adeline Roux-Langlois, François-Xavier Standaert

Venue: CHES 2026 (open access)

Summary: Established the computational hardness of the Learning with Physical Rounding (LWPR) problem in the presence of measurement noise by reducing it to the standard LWE assumption. Provided a comprehensive statistical analysis demonstrating that, despite a non-tight reduction, the results guarantee physical security properties comparable to those achieved through masking countermeasures.

Learning with Errors with Output Dependencies: LWE, LWR, and Physical Learning Problems Under the Same Umbrella

Authors: Clément Hoffmann, Pierrick Méaux, Mélissa Rossi, François-Xavier Standaert

Venue: PQCrypto 2026 (ePrint 2025/2225)

Summary: Introduced *LWE-OD*, a novel framework encompassing classical learning problems (LWE, LWR) and physical variants (LWPR). Established a formal simulator-based reduction to LWE, defining explicit hardness conditions for key-dependent error distributions. Demonstrated the framework's utility by deriving new proofs of known reductions for the security of LWR and LPN-OD.

Learning with Physical Rounding for Linear and Quadratic Leakage Functions

Authors: Clément Hoffmann, Pierrick Méaux, Charles Momin, Yann Rotella, François-Xavier Standaert, Balazs Udvarhelyi

Venue: CRYPTO 2023 (ePrint 2025/2226)

Summary: Investigated the security of *Learning with Physical Rounding (LWPR)* for fresh re-keying schemes. Demonstrated the vulnerability of serial implementations to worst-case leakage attacks while proving the inherent robustness of parallel architectures. Established necessary conditions for security based on leakage function Boolean properties.

When Bad News Become Good News: Towards Usable Instances of Learning with Physical Errors

Authors: Davide Bellizia, Clément Hoffmann, Dina Kamel, Pierrick Méaux, François-Xavier Standaert

Venue: CHES 2022 (open access)

Summary: Generalized hard physical learning problems to the Learning With Errors (LWE) setting (LWPE). Identified mathematical data dependencies in physical error distributions that could reduce security and developed design tweaks to make LWPE samples natively robust. Discussed the theoretical framework for the secure integration of inexact computing in LWE-based primitives.

Learning Parity with Physical Noise: Imperfections, Reductions and FPGA Prototype

Authors: Davide Bellizia, Clément Hoffmann, Dina Kamel, Hanlin Liu, Pierrick Méaux, François-Xavier Standaert, Yu Yu

Venue: CHES 2021 (open access)

Summary: Formalized the security of the Learning Parity with Physical Noise (LPPN) problem by providing a rigorous and tight reduction to the standard LPN problem. Proved that security is maintained despite physical data dependencies in error probabilities.

HYBRID HOMOMORPHIC ENCRYPTION

Secure and Efficient Transciphering for FHE-based MPC

Authors: Diego F. Aranha, Antonio Guimarães, Clément Hoffmann, Pierrick Méaux

Venue: CHES 2025 (ePrint 2024/1702)

Summary: Formalized the security framework for HHE ciphers against Related-Key Attacks (RKA) in Multi-Party Computation (MPC). Conducted the related-key security analysis of the Margrethe cipher, proving its native robustness in malicious multi-party environments. Quantified security bounds and demonstrated Margrethe's advantages over state-of-the-art ciphers for homomorphic evaluation.

The Patching Landscape of Elisabeth-4 and the Mixed Filter Permutator Paradigm

Authors: Clément Hoffmann, Pierrick Méaux, François-Xavier Standaert

Venue: Indocrypt 2023 (ePrint 2023/1895)

Summary: Addressed algebraic weaknesses in the Elisabeth-4 cipher by designing three secure successors: Elisabeth-b4, Gabriel, and Margrethe. Introduced the Mixed Filter Permutator paradigm to leverage large LUT evaluations in recent FHE schemes. Developed efficient algorithm to assess

the Boolean properties of filter functions with large input.

Towards Case-Optimized Hybrid Homomorphic Encryption: Featuring the Elisabeth Stream Cipher

Authors: Orel Cosseron, Clément Hoffmann, Pierrick Méaux, François-Xavier Standaert

Venue: Asiacrypt 2022 (ePrint 2022/180, code)

Summary: Introduced the Elisabeth stream cipher family for Hybrid Homomorphic Encryption (HHE). Conducted the symmetric security analysis and developed a high-performance Rust implementation. Demonstrated the scheme's efficiency by performing the first homomorphic neural network inference on HHE-encrypted data, surpassing the depth limitations of previous state-of-the-art HHE applications.

Transciphering, Using FiLIP and TFHE for an Efficient Delegation of Computation

Authors: Clément Hoffmann, Pierrick Méaux, Thomas Ricosset

Venue: Indocrypt 2020 (ePrint 2020/1373, code)

Summary: First practical implementation of the FiLIP stream cipher family using third-generation homomorphic encryption (TFHE). Developed novel strategies for the homomorphic evaluation of symmetric Boolean functions and XOR-threshold filters. Implemented and benchmarked TGSW/TLWE hybrid approaches, achieving significant speedups over state-of-the-art HElib implementations.

Preprints Under Review

Kettle: Short Post-Quantum Threshold Signatures from the HAWK Signature Scheme

Authors: Calvin Abou Haidar, Daniel Escudero, Thomas Espitau, Clément Hoffmann, Kaoru Takemure, Mehdi Tibouchi, Hernán Darío Vanegas Madrigal

Status: *Under review* (ePrint 2026/2288, code)

Summary: Identified the surprising MPC-friendliness of the HAWK signature scheme and built Kettle, a UC-secure threshold signing protocol that outputs specification-compliant HAWK signatures and scales to an arbitrary number of parties. Kettle offers essentially the shortest signature size among lattice-based threshold signatures, with as low as 2 online rounds and 2 kB of online communication per party, and comes with a proof-of-concept implementation in the MP-SPDZ framework.

Standardization Submissions

Kettle: Short Threshold HAWK Signatures

Submitters: Calvin Abou Haidar, Daniel Escudero, Thomas Espitau, Clément Hoffmann, Haruhisa Kosuge, Kaoru Takemure, Mehdi Tibouchi, Hernán Darío Vanegas Madrigal

Call: NIST Multi-Party Threshold Cryptography (MPTC) call, 2026 (preview document, preview talk)

Summary: Submission package of the Kettle threshold signing scheme, producing standard HAWK signatures, to the NIST call for multi-party threshold schemes. Presented it at the NIST Threshold Call Preview Talks (TCPT3).

Other Publications

Practical Leakage Attack on HAWK's Gaussian Sampler

Authors: Calvin Abou Haidar, Clément Hoffmann

Venue: SCIS 2026, January 2026 (national symposium, not peer-reviewed)

Summary: Developed a small practical leakage template attack targeting the HAWK post-quantum signature scheme. Exploited an implementation flaw in the Gaussian sampler and introduced a novel trick to bypass the *symbreak* operation, successfully recovering the full secret key on an ARM Cortex-M4 platform.

Presentations

CONFERENCE TALKS (SPEAKER)

2026: **CHES 2026**, Antalya, Türkiye — *Hardness of Learning with Physical Rounding and Noise from Learning With Errors*

2026: **CRYPTO 2026**, Santa Barbara, USA — *Maskaglia: A New, Efficient Approach to Masked Discrete Gaussian Sampling*

2026: **PQCrypto 2026**, Saint-Malo, France — *Learning with Errors with Output Dependencies*

2025: **CHES 2025**, Kuala Lumpur, Malaysia — *Secure and Efficient Transciphering for FHE-based MPC*

2023: **Indocrypt 2023**, Goa, India — *The Patching Landscape of Elisabeth-4 and the Mixed Filter Permutator Paradigm*

2023: **PKC 2023**, Atlanta, USA — *POLKA: Towards Leakage-Resistant Post-quantum CCA-Secure Public Key Encryption*

2022: **CHES 2022**, Leuven, Belgium — *When Bad News Become Good News: Towards Usable Instances of Learning with Physical Errors*

2021: **CHES 2021**, online — *Learning Parity with Physical Noise: Imperfections, Reductions and FPGA Prototype*

2020: **Indocrypt 2020**, online — *Transciphering, Using FiLIP and TFHE for an Efficient Delegation of Computation*

INVITED SEMINARS

2026: **Academia Sinica**, Taipei, Taiwan — *Maskaglia: A New, Efficient Approach to Masked Discrete Gaussian Sampling*

2026: **UCLouvain Crypto Group**, Louvain-la-Neuve, Belgium — *Maskaglia: A New, Efficient Approach to Masked Discrete Gaussian Sampling*

2025: **CWI**, Amsterdam, Netherlands — *Physical Learning Problems for Side-Channel Analysis*

2023: **University of Luxembourg**, Luxembourg — *Advances in Leakage-Resilient KEM*

OTHER TALKS & POSTERS

2026: **NIST Threshold Call Preview Talks (TCPT3)**, online — *Kettle: Short Threshold HAWK Signatures* (link)

2025: **CANS 2025** (poster), Osaka, Japan — *POLKA: A Practical Leakage-Resilient Post-Quantum PKE Scheme and its Leveled Implementation*